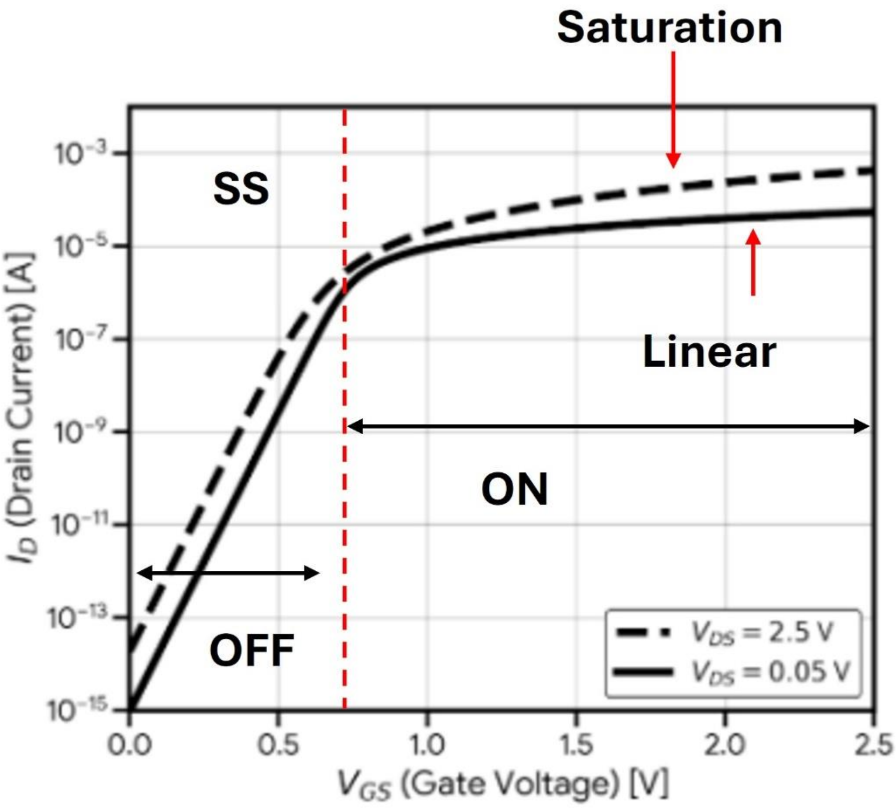
Q1. From the figure given below: (6 Marks)

a. Annotate the subthreshold region, linear and saturation region **(2 Marks)**



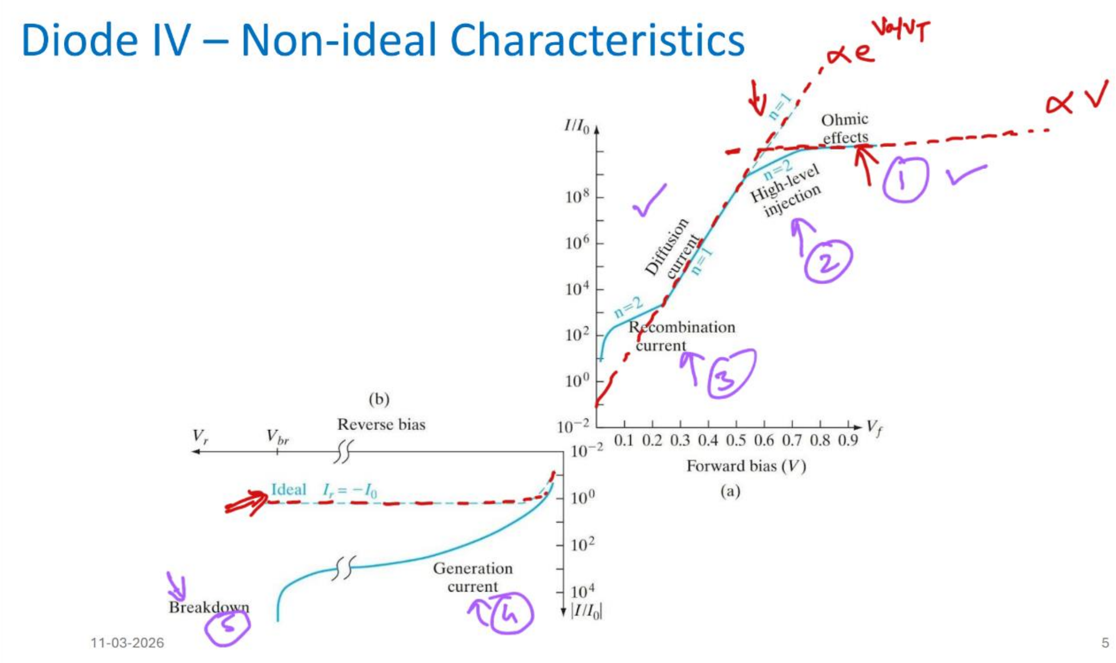
b. Which region can be used for switch operation i.e., ON and OFF region ? **(1 Marks)**
Sol:- Linear and OFF (SS) region are used. Note annotation of OFF and ON regions are required here

c. Which region can be used for amplifier operation? **(1 Marks)**
Sol :- Saturation region is used, here V_{ds} = 2.5 V shows saturation. Note annotation of OFF and saturation are required here .

d. What changes are observed by increasing the drain voltage to 2.5V? Explain the observations? **(2 Marks)**
Sol:- For OFF region (i.e $V_{gs} < V_{th}$) , Subthreshold Swing remains same. However, leakage increases. For ON region, the device is in Saturation as opposed to linear region in V_{ds} = 50 mV.

e. Calculate the SS? **(2 Marks)**
Sol :- ΔVgs = 0.50 b/w 1e-15 and 1e-9 showing 6 decades of change, gives us 0.5/6 = 83.3 mV/dec

Q2. For the non-ideal diode I-V shown in figure, name the different regions of operation. Reason each operation briefly. **(6 Marks)**



Diffusion or low-level injection :-Carrier transport dominated by concentration gradient, where injected carriers are the dominant current source or minority carrier recombination in QNR region.

High Level Injection :- Injected carrier concentration becomes comparable to or greater than the doping level, significantly altering both minority and majority carrier populations.

Recombination current:- Current resulting from electron–hole recombination (typically in the depletion region).

Generation current:- Current due to thermal generation of electron–hole pairs in the depletion region.

Series Resistance:- Ohmic resistance from bulk semiconductor that causes voltage drop and limits current at high bias.

Breakdown:-At high reverse bias, carriers gain enough energy to cause impact ionization, generating more electron–hole pairs and leading to a sudden large increase in current. For low doping, avalanche breakdown dominates.

Q3. For a MOSCAP with $SiO2$ as oxide and Si as semiconductor, a CV measurement is shown. Answer the following: **(8 Marks)**

a. Calculate the Oxide thickness? (1 Marks)

a) $C_{ox} = \frac{\varepsilon_{SiO_2} A}{t_{ox}}$; $\frac{C_{ox}}{A} = 3.4\times10^{-8}$ F/cm^2

$\Rightarrow t_{ox} = \frac{8.85\times10^{-14}\times 4}{3.4\times10^{-8}} = 104\times10^{-7}$ cm

$\boxed{t_{ox} \simeq 104 \text{ nm}}$

b. What is the type of the semiconductor? (1 Marks)

b) P-type as inversion can be seen on positive polarity

c. Calculate the doping of the semiconductor? (2 Marks)

c) Doping can be found from High Freq CV in inversion region where Cmin can be seen. Here, Cox & Cdep capacitance are in series

Cdep is result of max depletion width at $\psi_s = 2\phi_B$.

$$\frac{1}{C_{min}} = \frac{1}{C_{ox}} + \frac{1}{C_{dep}}$$

$C_{min} \simeq 8e-9$ F/cm^2
$C_{ox} \simeq 3.4e-8$ F/cm^2 (1 mark)

$\Rightarrow C_{dep} = 1.04\times10^{-8} = \frac{\varepsilon_{Si}}{W_{max}} = \frac{8.85\times10^{-14}\times12}{W_{max}}$

$\Rightarrow W_{max} = \sqrt{\frac{2\varepsilon_{Si}(2\phi_B)}{qN_A}} = 10^{-4}$ cm — (1)

$\Rightarrow \frac{\phi_B}{N_A} = 5.1\times10^{-16}$

(5 marks) $\phi_B = \frac{KT}{q}\ln(N_A/n_i) = 5.1\times10^{-16} N_A$

given $n_i = 1.1\times10^{10}$/cm^3

$$\Rightarrow N_A = 5.09\times 10^{13} \ln\left(N_A/n_i\right) \quad \text{--- (2)}$$

Solve iteratively.

Assume some value of N_A.

say $N_A = 10^{17}\ /cm^3$

Put in RHS of (2)

$\Rightarrow$ New $N_A = 8.16\times 10^{14}\ /cm^3$

(5 marks)

Put this new N_A again in RHS of (2)

$\Rightarrow$ New $N_A = 5.7\times 10^{14}\ /cm^3$

Put again in RHS of (2)

$\Rightarrow$ new $N_A = 5.5\times 10^{14}\ /cm^3$

So, N_A is $\approx \underline{5.5\times 10^{14}}\ /cm^3$

d. **Calculate the Threshold voltage V_{th} (Assume work function of metal and semiconductor are equal i.e. $\phi_m=\phi_{Si}$) (2 Marks)**

d) $V_T = V_{FB} + V_{ox} + \psi_s$

$V_{FB} = 0$ as given $\phi_m = \phi_{si}$

$\Rightarrow \quad V_{ox} = \dfrac{-Q_{dep}}{C_{ox}} = \dfrac{qN_A W_{max}}{C_{ox}}$

(1 mark)

from (1)

$W_{max} = 10^{-4}$ cm

$C_{ox} = 3.4 \times 10^{-8}$ F/cm² from part (a)

$N_A \simeq 5.5 \times 10^{14}$ /cm³ from part (c)

$\Rightarrow \quad V_{ox} = 0.26$ V

(1 mark)

$\psi_s = 2\phi_B$ (at inversion)

$= 2 V_T \ln(N_A/n_i)$;

$\psi_s = 0.56$ V

$\Rightarrow \quad \boxed{V_{T0} = 0.26 + 0.56 = 0.82\text{V}}$ — (3)

e. **Calculate the value of Flatband capacitance, Flatband Voltage and actual V_{th} using the same CV measurements. (2 Marks)**

Flat-band situation occurs when the bands are flat.

When band are flat the charge screening is given debye length.

$$\frac{1}{C_{FB}} = \frac{1}{C_{ox}} + \frac{1}{C_{debye}}$$

$$C_{debye} = \frac{\varepsilon_{si}}{L_{Debye}} = \frac{\varepsilon_{si}}{\sqrt{\frac{\varepsilon_{si} kT}{q^2 N_A}}}$$

$$C_{debye} = \sqrt{\frac{q^2 N_A \varepsilon_{si}}{kT}}$$

$$C_{debye} = 6 \times 10^{-8} \ F/cm^2$$

$$\Rightarrow \boxed{C_{FB} = 2.17 \times 10^{-8} \ F/cm^2}$$

from graph @ $C_{FB} = 2.17 \times 10^{-8}$ F/cm^2

(1 mark)

$$\Rightarrow \boxed{V = V_{FB} \simeq -0.8 \ V}$$

h)

$$V_T = V_{FB} + V_{T0}$$

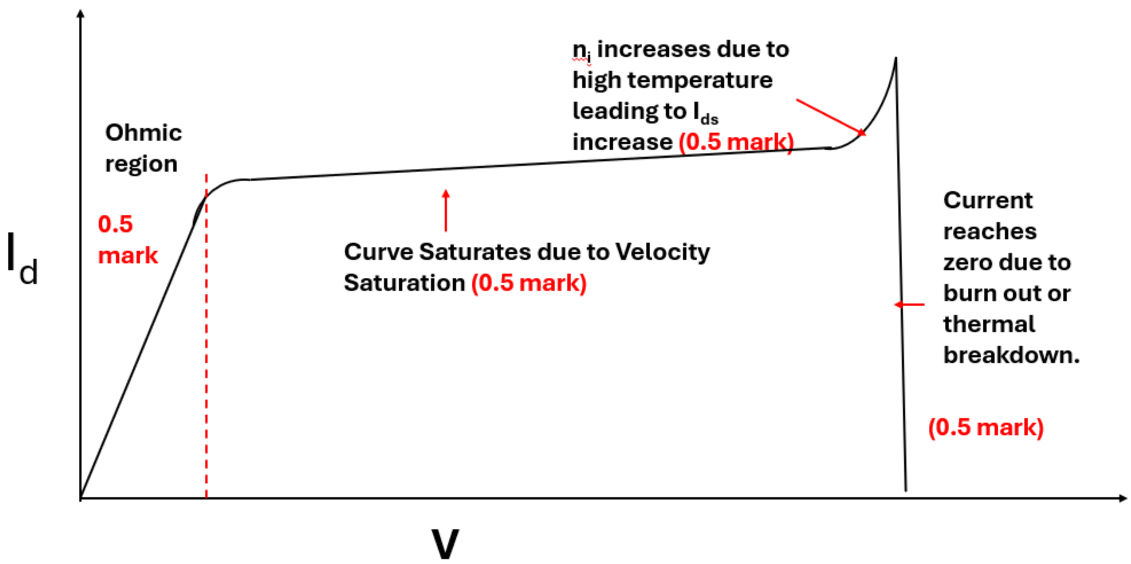
(1 mark)

Actual: $V_T = -0.8 + 0.82 = \underline{\underline{0.02 V}}$

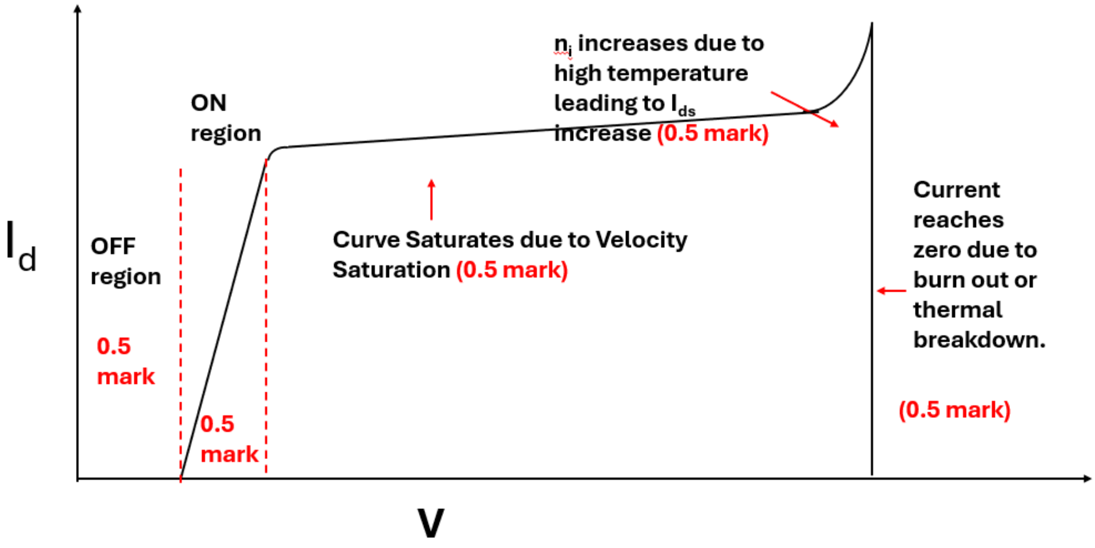
(The moscap is already in inversion due to V_{FB})

Q4. Answer the following questions: **(10 Marks)**

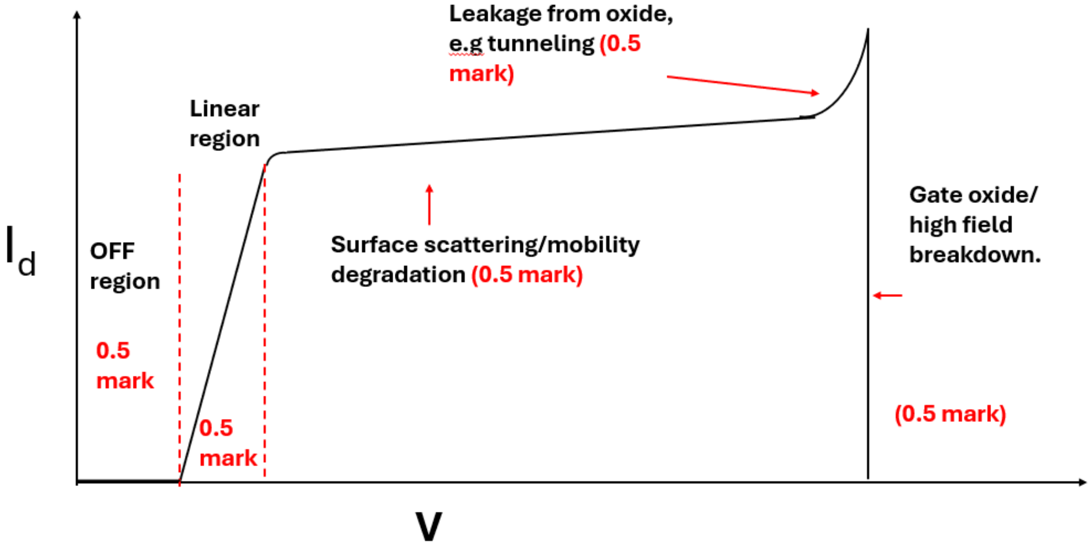
a. Plot I-V for a resistor. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V. **(2 Marks)**



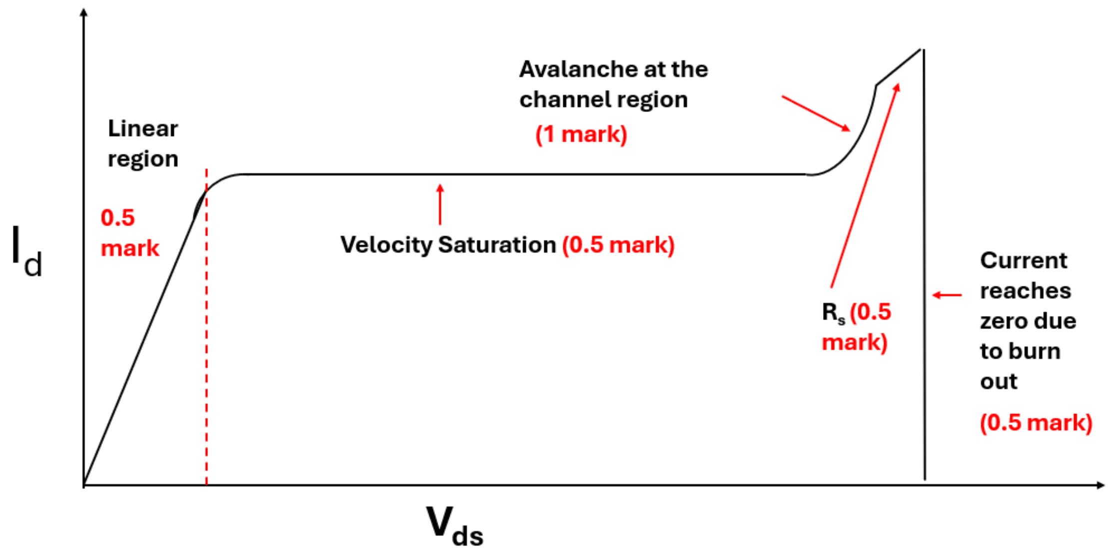
b. b. Plot I-V for a diode in a linear scale. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V. **(2 Marks)**



c. Plot I_{ds} vs V_{gs} plot in a linear scale with small Vds. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V.(**3 Marks)**



d. Plot Ids vs Vds plot in linear scale with $V_{gs} > V_{th}$. Keep increasing the voltage until the device burns out. Draw and explain the effect in different part of the I-V.(**3 Marks)**



Avalanche at the channel region
(1 mark)
Linear region
0.5 mark
I_d
Velocity Saturation (0.5 mark)
Current reaches zero due to burn out
(0.5 mark)
R_s (0.5 mark)
V_{ds}

Correct answer have been marked with blue colour

Q1. Subthreshold swing improves (decreases) when:

A. Cox ↓ B. $Cdep$ ↓ C. Temperature ↑ D. Ideality factor ↑

Q2. For p–n junctions with $NA = ND = 1015\ cm{-3}$, which material-based p-n junction diode loses rectifying behavior at the lowest temperature?

A. Ge B. Si C. GaAs D. All same

Q3. Velocity saturation in short-channel MOSFETs makes IDapproximately:

A. $\propto (VGS - VT)^2$ B. $\propto (VGS - VT)$ C. Independent of VGS D. $\propto VDS^2$

Q4. Compared to planar MOSFET, FinFET subthreshold slope is:

A. Higher B. Lower C. Same D. Zero

Q5. For a diode, if temperature increases, forward voltage at a fixed current:

A. Increases B. Decreases C. Remains constant D. First increases then decreases

Q6. An NMOS with VT = 0.6 V operates at VGS = 1.6 V. The boundary $VDS$for saturation is:

A. 0.6 V B. 1.0 V C. 1.6 V D. 2.2 V

Q7. For solar cell, increasing illumination shifts I–V curve:

A. Upward B. Downward C. Left D. No shift

Q8. Photodiode is operated in reverse bias to:

A. Reduce current B. Increase depletion width C. Increase recombination D. Reduce speed

Q9. Zener breakdown occurs due to:

A. Impact ionization B. Tunneling C. Diffusion D. Drift

Q10. Flash memory stores data in:

A. Resistance B. Floating gate C. Current D. Inductance